write a C program to simulate page replacement algorithms.
i. FIFO ii. LRU iii. Optimal.

```
#include<stdio.h>
#define MAX 100
int pages[MAX], FRAMES, PAGES;

int inframes(int frames[], int page){
    for(int i=0; i<FRAMES; i++)
        if(frames[i]==page) return 1;
    return 0;
}

void FIFO(){
    int frames[FRAMES];
    for(int i=0;i<FRAMES;i++)frames[i] = -1;

    int faults = 0, ptr = 0;
    printf("\n--FIFO--\n");
    for(int i=0; i<PAGES;i++){
        if(!inframes(frames,page[i])){
            frames[ptr] = pages[];
            ptr=(ptr+1)%FRAMES;
            faults++;
            printf("Page %d -> ", pages[i]);
            for(int j=0;=j<FRAMES;j++)
                printf("%d", frames[j]);
            printf("FAUIT\n");
        }else{
            printf("Page %d -> ", pages[i]);
            for(int j=0;j<FRAMES;j++)
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            printf("%d", frames[j]);
        printf("HIT\n");
    }
}
printf("Total faults: %d\n", faults);
}

void LRU(){
    int frames[FRAMES], lastUsed[FRAMES];
    for(int i=0; i<FRAMES; i++){
        frames[i] = -1;
        lastUsed[i] = -1;
    }
    int faults = 0;
    printf("\n--LRU--\n");
    for(int i=0; i<PAGES; i++){
        if(!inFrames(frames, pages[i])){
            int replace = 0;
            for(int j=0; j<FRAMES; j++){
                if(frames[j]==-1){
                    replace = j;
                    break;
                }
                if(lastUsed[j] < lastUsed[replace])
                    replace = j;
            }
            frames[replace] = pages[i];
            lastUsed[replace] = i;
            faults++;

            printf("Pages %d -> ", pages[i]);
```

```
        for(int j=0; j<FRAMES; j++)
            printf("%d", frames[j]);
        printf("FAULT\n");
    }else{
        for(int j=0; j<FRAMES; j++)
            if(frames[j]==pages[i])
                lastUsed[j]=i;
        printf("Pages%d -> ", pages[i]);
        for(int j=0; j<FRAMES; j++)
            printf("%d", frames[j]);
        printf("HIT\n");
    }
}
printf("Total Faults: %d\n", faults);
}
void Optimal(){
    int frames[FRAMES];
    for(int i=0; i<FRAMES; i++) frames[i]=-1;

    int faults=0, filled=0;

    printf("\n--OPTIMAL--\n");
    for(int i=0; j<PAGES; i++){
        if(!inFrames(frames,pages[i])){
            if(filled<FRAMES){
                frames[filled++]=pages[i];
            }else{
                int replace=0; farthest=-1;
                for(int j=0; j<FRAMES; j++){
                    int nextUse=PAGES;
```

```
            for(int k=j+1 ; k<PAGES; k++)
            {   if(pages[k]==frames[j]){
                    nextUse=k;
                    break;
                }
            }
            if(nextUse>farthest){
                farthest=nextUse;
                replace=j;
            }
        }
        frames[replace]=pages[i];
    }
    faults++;
    printf("Page %d -> ", pages[i]);
    for(int j=0; j<FRAMES; j++)
        printf("%d", frames[j]);
    printf("FAULT\n");
}else{
    printf("Page %d ->", pages[i]);
    for(int j=0; j<FRAMES ; j++)
        printf("%d", frames[j]);
    printf("HIT\n");
    }
}
printf("Total faults: %d\n", faults);
}

int main(){
    printf("Enter number of frames: ");
    scanf("%d", &FRAMES);
```

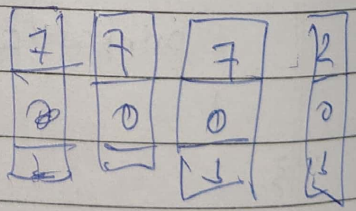
```
printf(" Enter number of pages :");
scanf("%d",&PAGES);

printf("Enter page reference string:\n");
for(int i=0; i<PAGES; i++){
    scanf("%d",&pages[i]);
}
FIFO();
LRU();
Optimal();
return 0;
}
```

output:

Enter number of pages : 20
Enter page reference string:
7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1
Enter number of frames : 3.

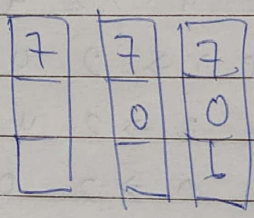
--FIFO--

Page 7 → 7 -1 -1 FAULT
Page 0 → 7 0 -1 FAULT
Page 1 → 7 0 1 FAULT
Page 2 → 2 0 1 FAULT
Page 0 → 2 0 1 ~~MISS~~ HIT
Page 3 → 2 3 1 FAULT
Page 0 → 2 3 0 FAULT
Page 4 → 4 3 0 FAULT
Page 2 → 4 2 0 FAULT
Page 3 → 4 2 3 FAULT



Page 0 → 0 2 3 FAULT
Page 3 → 0 2 3 HIT
Page 2 → 0 2 3 HIT
Page 1 → 0 1 3 FAULT
Page 2 → 0 1 2 FAULT
Page 0 → 0 1 2 HIT
Page 1 → 0 1 2 HIT
Page 7 → 7 1 2 FAULT
Page 0 → 7 0 2 FAULT
Page 1 → 7 0 1 FAULT
Total page faults: 15

--LRU--



Page 7 → 7 -1 -1 FAULT
Page 0 → 7 0 -1 FAULT
Page 1 → 7 0 1 FAULT
Page 2 → 2 0 1 FAULT
Page 0 → 2 0 1 HIT
Page 3 → 2 0 3 FAULT
Page 0 → 2 0 3 HIT
Page 4 → 4 0 3 FAULT
Page 2 → 4 0 2 FAULT
Page 3 → 4 3 2 FAULT
Page 0 → 0 3 2 FAULT
Page 3 → 0 3 2 HIT
Page 2 → 0 3 2 HIT
Page 1 → 1 3 2 FAULT
Page 2 → 1 3 2 HIT
Page 0 → 1 0 2 FAULT
Page 1 → 1 0 2 HIT
Page 7 → 7 0 2 FAULT

Page 0 → 1 0 7 HIT
Page 1 → 1 0 7 HIT
Total Page faults = 12.

--OPTIMAL--

Page 7 →	7	-1	-1	FAULT
Page 0 →	7	0	-1	FAULT
Page 1 →	7	0	1	FAULT.
Page 2 →	2	0	1	FAULT
Page 0 →	2	0	1	HIT
Page 3 →	2	0	3	~~HI~~ FAULT
Page 0 →	2	0	3	HIT
Page 4 →	2	4	3	FAULT
Page 2 →	2	4	3	HIT
Page 3 →	2	4	3	HIT
Page 0 →	2	0	3	FAULT
Page 3 →	2	0	3	HIT
Page 2 →	2	0	3	HIT
Page 1 →	2	0	1	FAULT
Page 2 →	2	0	1	HIT
Page 0 →	2	0	1	HIT
Page 1 →	2	0	1	HIT
Page 7 →	7	0	1	FAULT
Page 0 →	7	0	1	HIT
Page 1 →	7	0	1	HIT

Total page faults = 9.

7	7	7
	0	0
		1

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